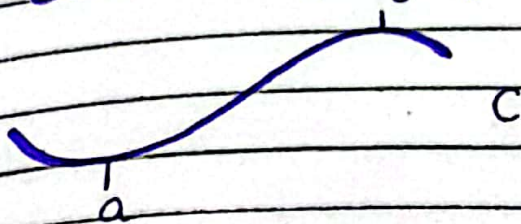


[1502]

line Integral



$$C: r(t) = x(t)i + y(t)j$$

$$\text{line Integral} := \int_C f(x,y) ds$$

formula:-

$$\int_C (f(x,y)) ds = \int_a^b f(x(t), y(t)) |r'(t)| dt$$

Example 1

Use the given parameterization, to evaluate the integral

$$\int_C (1 + xy^2) ds$$

$$x = t$$

$$y = 2t$$

$$C: r(t) = \underbrace{t}_x i + \underbrace{2t}_y j \quad (0 \leq t \leq 1)$$

$$r'(t) = i + 2j$$

aka
derivative
w.r.t t

$$|r'(t)| = \sqrt{1^2 + 2^2}$$

$$= \sqrt{5}$$

$$\int_C (1 + xy^2) \, ds$$

$$= \int_0^1 (1 + t(2t)^2) |r'(t)| \, dt$$

$$= \int_0^1 (1 + 4t^3) \sqrt{5} \, dt$$

$$= \sqrt{5} \int_0^1 (1 + 4t^3) \, dt$$

$$= \sqrt{5} \left[t + 4 \frac{t^4}{4} \right]_0^1$$

$$= \sqrt{5} [t + t^4]_0^1$$

$$= \sqrt{5} [(1-0) + (1-0)]$$

$$= \sqrt{5} [1+1]$$

$$= \sqrt{5} (2)$$

$$= 2\sqrt{5}$$

$$b) \quad C: r(t) = (1-t)i + (2-2t)j$$

$$0 \leq t \leq 1$$

$$x = 1-t$$

$$y = 2-2t$$

$$\begin{aligned} r'(t) &= i - ti + 2j - 2tj \\ &= 0 - i + 0 - 2j \\ &= -i - 2j \end{aligned}$$

$$\begin{aligned} |r'(t)| &= \sqrt{(-1)^2 + (-2)^2} \\ &= \sqrt{5} \end{aligned}$$

$$\int_C (1+xy^2) \, ds$$

$$= \sqrt{5} \int_0^1 (1 + (1-t)(2-2t)^2) \, dt$$

$$\sqrt{5} \int_0^1 (1 + (1-t)2^2(1-t)^2) \, dt$$

$$= \sqrt{5} \int_0^1 (1 + 2^2(1-t)^3) \, dt$$

$$= \sqrt{5} \int_0^1 (1 + 4(1-t)^3) \, dt$$

$$= \sqrt{5} \int_0^1 (1 - 4(-(1-t)^3)) \, dt$$

$$\sqrt{5} \left[t - \frac{4(1-t)^4}{4} \right]_0^1$$

$$= \sqrt{5} \left(t - (1-t)^4 \right) \Big|_0^1$$

$$\text{I.e.} \quad \sqrt{5} \left(\left[1 - (1-1)^4 \right] - \left[0 - (1-0)^4 \right] \right)$$

$$= \sqrt{5} \left(\left[1 - 0 \right] - \left[0 - 1 \right] \right)$$

$$= \sqrt{5} \left(1 - 0 - 0 + 1 \right)$$

$$= \sqrt{5} (2)$$

$$= 2\sqrt{5}$$

2nd formula for line Integral

$$\int_a^b f(x, y) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Q

Example 2 Evaluate the line Integral

$$\int_C (xy + z^2) ds$$

along the helix

$$x = \cos t, \quad y = \sin t, \quad z = t$$

$$(0 \leq t \leq \pi)$$

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t, \quad \frac{dz}{dt} = 1$$

$$\sqrt{\left[\frac{dx}{dt}\right]^2 + \left[\frac{dy}{dt}\right]^2 + \left[\frac{dz}{dt}\right]^2}$$

$$= \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2}$$

$$= \sqrt{\sin^2 t + \cos^2 t + 1}$$

$$= \sqrt{1+1} = \sqrt{2}$$

$$\therefore \int_C (xy + z^2) ds$$

$$= \sqrt{2} \int_0^{\pi} (\cos t \cdot \sin t + t^2) dt$$

$$= \sqrt{2} \left[\frac{\sin^2 t}{2} + \frac{t^3}{3} \right]_0^{\pi}$$

$$= \sqrt{2} \left(\frac{\sin^2 \pi}{2} + \frac{\pi^3}{3} - \frac{\sin^2 0}{2} + \frac{0^3}{3} \right)$$

$$= \sqrt{2} \left[\frac{\pi^3}{3} \right]$$

$$= \sqrt{2} \frac{\pi^3}{3}$$

$$\underline{Q19)} \int_C \frac{1}{1+x} ds$$

$$C: r(t) = ti + \frac{2}{3} t^{3/2} j$$

(0 → 3)

$$x = t$$

$$r'(t) = i + \frac{3}{2} \cdot \frac{2}{3} t^{1/2} j$$

$$= i + t^{1/2} j$$

$$|r'(t)| = \sqrt{(1)^2 + (t^{1/2})^2}$$

$$= \sqrt{1+t}$$

$$= \int_0^3 \frac{1}{1+t} \cdot \sqrt{1+t} dt$$

$$= \int_0^3 (1+t)^{-1/2} dt$$

$$= \left[\frac{(1+t)^{1/2}}{1/2} \right]_0^3$$

$$= \left[2(1+t)^{1/2} \right]_0^3$$

$$= [2(1+3)^{\frac{1}{2}}] - [2(1+0)^{\frac{1}{2}}]$$

$$= 4 - 2$$

$$= 2$$

3rd Method to Solve the line Integral

1.1

$$\int_C f(x,y) dx + g(x,y) dy$$
$$x = x(t), \quad y = y(t)$$

Example 7

Evaluate

$$\int (3x^2 + y^2) \cdot dx + 2xy \cdot dy$$

along circular arc C

$$x = \cos t, \quad y = \sin t \quad (0 \leq t \leq \pi/2)$$

$$dx = -\sin t \, dt$$

$$dy = \cos t \, dt$$

$$= \int_0^{\pi/2} (3\cos^2 t + \sin^2 t) \cdot (-\sin t \, dt) + 2(\cos t)(\sin t)(\cos t \, dt)$$

$$= \int_0^{\pi/2} (-3\cos^2 t \sin t - \sin^3 t + 2\cos^2 t \sin t) \, dt$$

$$= \int_0^{\pi/2} (-\cos^2 t \sin t - \sin^3 t) \, dt$$

Example 8

line Integral

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int F(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

$$C: \mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$$

Example:

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$

$$\mathbf{F}(x, y) = \cos x \mathbf{i} + \sin x \mathbf{j}$$

$$a) C: \mathbf{r}(t) = -\frac{\pi}{2}\mathbf{i} + t\mathbf{j} \quad (1 \leq t \leq 2)$$

$$x = -\frac{\pi}{2}, \quad y = t$$

$$\begin{aligned} \mathbf{F}(\mathbf{r}(t)) &= \cos\left[-\frac{\pi}{2}\right]\mathbf{i} + \sin\left[-\frac{\pi}{2}\right]\mathbf{j} = -\cos(90) - \mathbf{j} \\ &= 0 - 1\mathbf{j} = -\mathbf{j} \quad \because \sin(-x) = -\sin x \end{aligned}$$

$$\mathbf{r}'(t) = 0 + \mathbf{j}$$

Apply to formula

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

$$= \int_1^2 -j \cdot j \, dt$$

$$= -1 \int_1^2 dt$$

$$= -1 [t]_1^2$$

$$= -1 (2-1)$$

$$= -1$$

$$j \cdot j = -1$$

b) $C: r(t) = ti + t^2j \quad (-1 \leq t \leq 2)$

$$x = t$$

$$y = t^2$$

$$T(r(t)) = \cos t i + \sin t j$$

$$r'(t) = i + 2tj$$

$$= \int_{-1}^2 (\cos t i + \sin t j) \cdot (i + 2tj) \, dt$$

$$i \cdot i = 1$$

$$= \int_{-1}^2 (\cos t + 2t \sin t) \, dt$$

$$= \int_{-1}^2 \cos t \, dt + 2 \int_{-1}^2 t \sin t \, dt$$

$$[\sin t]_{-1}^2 + 2 \left[t \int \sin t \, dt - \int \frac{d}{dt}(t) \int \sin t \, dt \, dt \right]$$

$$(\sin 2 + \sin 1) + 2 \left[t(-\cos t) - \int 1(-\cos t) \, dt \right]$$

$$= \sin 2 + \sin 1 + 2 \left[t(-\cos t) + \sin t \right]_{-1}^2$$

$$= \sin 2 + \sin 1 + 2 \left[-t \cos t + \sin t \right]_{-1}^2$$

$$= \sin 2 + \sin 1 + 2 \left[-2 \cos 2 + \sin 2 - (-1 \cos(-1) + \sin(-1)) \right]$$

$$= \sin 2 + \sin 1 + 2 \left[-2 \cos 2 + \sin 2 + \cos(-1) - \sin(-1) \right]$$

$$= \sin 2 + \sin 1 - 4 \cos 2 + 2 \sin 2 + 2 \cos(-1) - 2 \sin(-1)$$

$$= 5.6 \dots$$

Q38)

$$\int_C F \cdot dx$$

$$F(x, y) = x^2 y i + 4j$$

$$C: r(t) = e^t i + e^{-t} j \quad (0 \leq t \leq 1)$$

$$x = e^t$$

$$y = e^{-t}$$

$$F(r(t)) = (e^t)^2 (e^{-t}) i + 4j$$

$$r'(t) = e^t i - e^{-t} j$$

$$\int_C F \cdot dx = \int_0^1 (e^t i + 4j) \cdot (e^t i - e^{-t} j) dt$$

$$= \int_0^1 (e^{2t} - 4e^{-t}) dt$$

$$= \left[\frac{e^{2t}}{2} - \frac{4e^{-t}}{-1} \right]_0^1$$

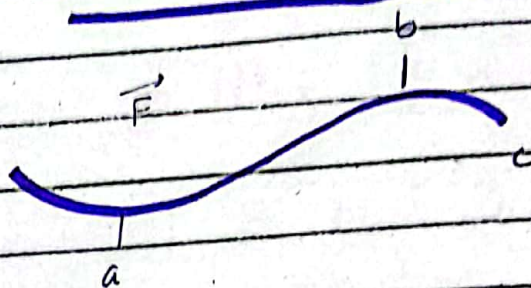
$$= \left[\frac{e^{2t}}{2} + 4e^{-t} \right]_0^1$$

$$= \frac{e^{2(1)}}{2} + 4e^{-1} - \left(\frac{e^{2(0)}}{2} + 4e^0 \right)$$

$$= \frac{e^2}{2} + \frac{4}{e} - \left(\frac{1}{2} + 4 \right)$$

$$= \frac{e^2}{2} + \frac{4}{e} - \frac{1}{2} - 4$$

WORK DONE



$$\text{Work done} = \int_C \mathbf{F} \cdot d\mathbf{r}$$

Q46) find the work done by force $\mathbf{F}$ on the particle along curve C

$$\mathbf{F}(x, y) = (x^2 + xy)\mathbf{i} + (y - x^2y)\mathbf{j}$$

$$C: x = t, \quad y = \frac{1}{t} \quad (1 \leq t \leq 3)$$

$$\mathbf{r}(t) = t\mathbf{i} + \frac{1}{t}\mathbf{j}$$

$$\mathbf{F}(\mathbf{r}(t)) = \left(t^2 + t\left[\frac{1}{t}\right]\right)\mathbf{i} + \left(\frac{1}{t} - t^2\left[\frac{1}{t}\right]\right)\mathbf{j}$$

$$= (t^2 + 1)\mathbf{i} + \left(\frac{1}{t} - t\right)\mathbf{j}$$

$$= (t^2 + 1)\mathbf{i} + \left[\frac{1-t^2}{t}\right]\mathbf{j}$$

$$\mathbf{r}'(t) = \mathbf{i} - \frac{1}{t^2}\mathbf{j}$$

$$t^{-3}$$

$$\frac{t^{-2}}{-2}$$

$$\int_c^b F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt$$
$$= \int_1^3 \left((t^2+1)i + \left(\frac{1-t^2}{t}\right)j \right) \cdot \left(i - \frac{1}{t^2}j \right) dt$$

$$= \int_1^3 \left((t^2+1) - \frac{1}{t^2} \left(\frac{1-t^2}{t} \right) \right) dt$$

$$= \int_1^3 \left[(t^2+1) - \frac{1+t^2}{t^3} \right] dt$$

$$= \int_1^3 \left[t^3 + 1 - \frac{1}{t^3} + \frac{t^2}{t^3} \right] dt$$

$$= \int_1^3 \left[t^3 + 1 - \frac{1}{t^3} + \frac{1}{t} \right] dt$$

$$= \left[\frac{t^4}{4} + t - \frac{t^{-2}}{-2} - \ln t \right]_1^3$$

$$= \left[\frac{t^4}{4} + t + \frac{t^{-2}}{2} - \ln t \right]_1^3$$

$$= \left[\frac{3^4}{4} + 3 + \frac{(3)^{-2}}{2} - \ln 3 \right] - \left[\frac{1}{4} + 1 + \frac{1^{-2}}{2} - \ln 1 \right]$$

$$= \frac{92}{4} - \ln 3$$

$$\underline{Q20)} \int_C \frac{x}{1+y^2} ds$$

$$C: x = 1 + 2t, y = t \quad (0 \leq t \leq 1)$$

$$r'(t) = 2i + j$$

$$|r'(t)| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$= \int_C \frac{x}{1+y^2} ds$$

$$= \int_0^1 \frac{1+2t}{1+t^2} \cdot \sqrt{5} dt$$

$$= \sqrt{5} \int_0^1 \frac{1}{1+t^2} dt + \sqrt{5} \int_0^1 \frac{2t}{1+t^2} dt$$

$$= \sqrt{5} \left[\tan^{-1} t \right]_0^1 + \sqrt{5} \left[\ln(1+t^2) \right]_0^1$$

$$= \sqrt{5} \left[\tan^{-1}(1) - \tan^{-1}(0) \right] + \sqrt{5} \left[\ln(1+1) - \ln(1+0) \right]$$

$$= \sqrt{5} \left[\frac{\pi}{4} - 0 \right] + \sqrt{5} \left[\ln 2 \right]$$

$$= \sqrt{5} \left(\frac{\pi}{4} \right) + \sqrt{5} (\ln 2)$$

$$= \sqrt{5} \left(\frac{\pi}{4} + \ln 2 \right)$$

$$Q21) \int 3x^2 y z \, ds$$

$$C: x=t, y=t^2, z=\frac{2}{3}t^3 \quad (0 \leq t \leq 1)$$

$$r'(t) = i + 2tj + 2t^2k$$

$$|r'(t)| = \sqrt{(1)^2 + (2t)^2 + (2t^2)^2}$$

$$= \sqrt{1 + 4t^2 + 4t^4} = \sqrt{(1 + 2t^2)^2}$$

$$\{ |r'(t)| = 1 + (2t^2) \}$$

$$\int_0^1 3t^2 \cdot t^2 \cdot \frac{2t^3}{3} \cdot (1 + 2t^2) \, dt$$

$$= \int_0^1 2t^7 \cdot (1 + 2t^2) \, dt$$

$$= \int_0^1 2t^7 + 4t^9 \, dt$$

$$= 2 \int_0^1 t^7 \, dt + 4 \int_0^1 t^9 \, dt$$

$$= 2 \left[\frac{t^8}{8} \right]_0^1 + 4 \left[\frac{t^{10}}{10} \right]_0^1$$

$$= \frac{1}{4} [t^8]_0^1 + \frac{2}{5} [t^{10}]_0^1$$

$$= \frac{1}{4} + \frac{2}{5} = \frac{13}{20}$$

$$Q22) \int_C \frac{e^{-z}}{x^2+y^2} dz$$

$$C: r(t) = 2\cos t i + 2\sin t j + t k \\ (0 \leq t \leq 2\pi)$$

$$r'(t) = -2\sin t i + 2\cos t j + k$$

$$|r'(t)| = \sqrt{(-2\sin t)^2 + (2\cos t)^2 + (1)^2}$$

$$= \sqrt{4\sin^2 t + 4\cos^2 t + 1} \\ = \sqrt{4(1) + 1} \\ = \sqrt{4+1} \\ = \sqrt{5}$$

$$\int_0^{2\pi} \frac{e^{-t}}{(2\cos t)^2 + (2\sin t)^2} \sqrt{5} dt$$

$$= \sqrt{5} \int_0^{2\pi} \frac{e^{-t}}{4\cos^2 t + 4\sin^2 t} dt$$

$$= \frac{\sqrt{5}}{4} \int_0^{2\pi} e^{-t} dt$$

$$= -\frac{\sqrt{5}}{4} [e^{-t}]_0^{2\pi}$$

$$= -\frac{\sqrt{5}}{4} [e^{-2\pi} - e^0]$$

$$= -\frac{\sqrt{5}}{4} (e^{-2\pi} - 1)$$

~~Q22~~ ~~PA~~

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}, \quad \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$f(x)$

$g(y)$

23) $\int_C (x+2y) dx + (x-y) dy$

$C: x = 2\cos t, y = 2\sin t \quad (0 \leq t \leq \pi/4)$

$dx = -2\sin t dt, dy = 2\cos t dt$

$$\int_0^{\pi/4} (2\cos t + 2(2\sin t)) \cdot (-2\sin t dt) + (2\cos t - 2\sin t) (2\cos t dt)$$

$$\int_0^{\pi/4} (2\cos t + 4\sin t) (-2\sin t dt) + (2\cos t - 2\sin t) (2\cos t dt)$$

$$= \int_0^{\pi/4} -4\cos t \sin t - 16\sin^2 t + 8\cos^2 t - 16\cos t \sin t dt$$

$$= \int_0^{\pi/4} 8\cos^2 t - 16\sin^2 t - 20\cos t \sin t dt$$

$$= \int_0^{\pi/4} (8\cos^2 t - 16 + 16\cos^2 t) dt - 20 \int_0^{\pi/4} \cos t \sin t dt$$

$$Q24) \int (x^2 - y^2) dx + x dy$$

$$C: x = t^{2/3}, \quad y = t \quad (-1 \leq t \leq 1)$$

$$dx = \frac{2}{3} t^{-1/3} dt, \quad dy = 1 dt$$

$$= \int_{-1}^1 ((t^{2/3})^2 - t^2) \left(\frac{2}{3} t^{-1/3} dt\right) + (t^{2/3})(1 dt)$$

$$= \int_{-1}^1 \left(\frac{2}{3} t^{4/3 - 1/3} - \frac{2t^{2 - 1/3}}{3} \right) dt + t^{2/3} dt$$

$$= \int_{-1}^1 \left(\frac{2}{3} t - \frac{2}{3} t^{5/3} \right) dt + t^{2/3} dt$$

$$= \int_{-1}^1 \frac{2}{3} (t - t^{5/3}) dt + t^{2/3} dt$$

$$= \frac{2}{3} \int_{-1}^1 t - t^{5/3} dt + \int_{-1}^1 t^{2/3} dt$$

$$= \frac{2}{3} \int_{-1}^1 (t - t^{5/3}) dt + \int_{-1}^1 (t^{2/3}) dt$$

$$= \frac{2}{3} \left[\frac{t^2}{2} - \frac{t^{8/3}}{8/3} \right]_{-1}^1 + \left[\frac{t^{5/3}}{5/3} \right]_{-1}^1$$

$$= \frac{2}{3} \left[\frac{t^2}{2} - \frac{3t^{8/3}}{8} \right]_{-1}^1 + \frac{3}{5} \left[t^{5/3} \right]_{-1}^1$$

$$\frac{2}{3} \left[\frac{t^2}{2} \right]_{-1}^1 - \frac{2}{3} \left[\frac{3t^{8/3}}{7} \right]_{-1}^1 + \frac{3}{5} \left[t^{5/3} \right]_{-1}^1$$

$$\frac{1}{3} \left[t^2 \right]_{-1}^1 - \frac{2}{8} \left[8t^{8/3} \right]_{-1}^1 + \frac{3}{5} \left[t^{5/3} \right]_{-1}^1$$

$$\frac{1}{3} [1-1] - \frac{2}{8} [t^{8/3}]_{-1}^1 + \frac{3}{5} [t^{5/3}]_{-1}^1$$

$$= \frac{1}{3} [1-1] - \frac{1}{4} [1 - (-1)^{8/3}] + \frac{3}{5} [1 - (-1)^{5/3}]$$

$$= 0 - \frac{1}{4} [1-1] + \frac{3}{5} [1+1]$$

$$= 0 - 0 + \frac{6}{5} = \frac{6}{5}$$

$$Q29) \int yz \, dx - xz \, dy + xy \, dz$$

$$\therefore x = e^t, y = e^{3t}, z = e^{-t}$$

$$(0 \leq t \leq 1)$$

$$dx = e^t \, dt, dy = 3e^{3t} \, dt, dz = -e^{-t} \, dt$$

$$= \int_0^1 (e^{3t} \cdot e^{-t})(e^t \, dt) - (e^t \cdot e^{-t})(3e^{3t} \, dt) + (e^t \cdot e^{3t})(-e^{-t} \, dt)$$

$$= \int_0^1 (e^{2t})(e^t \, dt) - (e^0)(3e^{3t} \, dt) + (e^{4t})(-e^{-t} \, dt)$$

$$= \int_0^1 (e^{3t} - 3e^{3t} - e^{3t}) \, dt$$

$$= \int_0^1 -3e^{3t} \, dt$$

$$= -3 \int_0^1 e^{3t} \, dt$$

$$= -3 \left[\frac{e^{3t}}{3} \right]_0^1$$

$$= - \left[e^{3t} \right]_0^1$$

$$= - \left[e^{3(1)} - e^0 \right] = - (e^3 - 1)$$

$$= 1 - e^3$$

$$Q30) \int x^2 dx + xy dy + z^2 dz$$

$$C: x = \sin t, y = \cos t, z = t^2$$

$$(0 \leq t \leq \pi/2)$$

$$dx = \cos t dt, dy = -\sin t dt, dz = 2t dt$$

$$= \int_0^{\pi/2} (\sin t)^2 (\cos t dt) + (\sin t \cos t) (-\sin t dt) + (t^2)^2 (2t dt)$$

$$= \int_0^{\pi/2} \sin^2 t \cos t dt - \sin^2 t \cos t dt + 2t^5 dt$$

$$= \int_0^{\pi/2} \sin^2 t \cos t - \sin^2 t \cos t dt + 2t^5 dt$$

$$= \int_0^{\pi/2} 2t^5 dt$$

$$= 2 \int_0^{\pi/2} t^5 dt$$

$$= 2 \left[\frac{t^6}{6} \right]_0^{\pi/2}$$

$$= \frac{1}{3} \left(\left[\frac{\pi}{2} \right]^6 - 0^6 \right)$$

$$= \frac{\pi^6}{192}$$

Q37) $F(x,y) = x^2 i + xy j$

C: $r(t) = 2\cos t i + 2\sin t j \quad (0 \leq t \leq \pi)$

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt$$

C: $r(t) = x(t)i + y(t)j$

Solution:

$$r'(t) = -2\sin t i + 2\cos t j$$

$$F(r(t)) = 4\cos^2 t i + 4\cos t \sin t j$$

$$\int_C F \cdot dr = \int_0^\pi (4\cos^2 t i + 4\cos t \sin t j) \cdot (-2\sin t i + 2\cos t j) dt$$

$$= \int_0^\pi (-8\cos^2 t \sin t + 8\cos^2 t \sin t) dt$$

$$= \int_0^\pi 0 dt$$

$$= 0$$

QNO. 40)

$$F(x, y, z) = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$$

$$C: (r(t)) = \sin t \mathbf{i} + 3 \sin t \mathbf{j} + \sin^2 t \mathbf{k} \\ (0 \leq t \leq \pi/2)$$

$$r'(t) = \cos t \mathbf{i} + 3 \cos t \mathbf{j} + 2 \sin t \cos t \mathbf{k}$$

$$F(r(t)) = \sin^2 t \mathbf{i} + \sin t \mathbf{j} + 3 \sin t \mathbf{k}$$

$$= \int_0^{\pi/2} (\sin^2 t \mathbf{i} + \sin t \mathbf{j} + 3 \sin t \mathbf{k}) (\cos t \mathbf{i} + 3 \cos t \mathbf{j} + 2 \sin t \cos t \mathbf{k}) dt$$

$$= \int_0^{\pi/2} (\sin^2 t \cos t + 3 \cos t \sin t + 6 \sin^2 t \cos t) dt$$

$$= \int_0^{\pi/2} (7 \sin^2 t \cos t + 3 \cos t \sin t) dt$$

Q No. 47)

$$F(x, y, z) = xyi + yzj + xzk$$

$$C: r(t) = ti + t^2j + t^3k$$

$$(0 \leq t \leq 1)$$

$$r'(t) = i + 2tj + 3t^2k$$

$$F(r(t)) = t^3i + t^5j + t^4k$$

$$\int_0^1 (t^3i + t^5j + t^4k) (i + 2tj + 3t^2k)$$

$$= \int_0^1 (t^3 + 2t^6 + 3t^6) dt$$

$$= \left[\frac{t^4}{4} + 2\frac{t^7}{7} + 3\frac{t^7}{7} \right]_0^1$$

$$= \frac{1}{4} + \frac{2}{7} + \frac{3}{7}$$

$$= \frac{27}{28}$$

$$Q39) F(x,y) = (x^2+y^2)^{-3/2} (xi+yj)$$

$$C: r(t) = e^t \sin t i + e^t \cos t j$$

$$(0 \leq t \leq 1)$$

$$x'(t) =$$